INCH-POUND

MIL-L-52413B(ME)
9 May 1990
SUPERSEDING
MIL-L-52413A(ME)
12 August 1968

MILITARY SPECIFICATION

LIFE PRESERVER, YOKE: ORAL INFLATION, TRAPPED AIR TYPE, DESIGN 9008

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adult-size, trapped-air type life preserver.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4220
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STANDARDS

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

TA13216E5045	- Life Preserver, Trapped Air Type, Design 9008.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FMT, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 413	- Rubber Property - Adhesion to Flexible Substrate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document

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takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The life preserver shall be in accordance with top assembly TA13216E5045 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the life preserver. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. The materials, except for the metallic parts, shall have been manufactured within 12 months prior to the date of delivery of the life preserver.

3.3.1 Material deterioration prevention and control. The life preserver shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the life preserver may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the life preserver may be newly fabricated from recovered materials to the maximum extent practicable, provided the life preserver produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the life preserver.

3.4 Fabrication. The coated side of the cloth shall be to the inside of the life preserver. The spacer insert pockets shall separate the inside of the life preserver into individual air compartments. Pockets shall be fabricated with the coated side of the cloth to the inside of the pocket, and shall be open to air compartments at each end. Tape used to reinforce cemented seams shall have the coated side of the cloth cemented to sections forming the seam.

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3.4.1 Cemented joints. Surfaces to be joined by cementing shall be clean and free of all foreign matter prior to the application of adhesive. Adhesive shall be evenly applied to each surface and the surfaces joined. All entrapped air and wrinkles shall be removed by rolling or stretching. The minimum strength for cemented joints shall be 7.5 pounds per inch of width where coated side of cloth is joined to coated side of cloth and 5 pounds per inch of width where uncoated side of cloth is joined to coated side of cloth.

3.4.2 Stitching. Ends of stitching that are not caught in other seams or stitching shall be backstitched a minimum of 0.5 inch. Thread breaks, skips, or run-offs shall be overstitched a minimum of 0.5 inch. Thread tension shall be maintained to prevent loose stitching and to embed the lock stitches in the material.

3.5 Oral inflation tube and valve. The oral inflation tube shall be so attached to the life preserver and the oral inflation valve so attached to the tube that there shall be no indication of separation when tested as specified in 4.5.2.1.1.

3.6 Weight. The weight of the complete life preserver shall not exceed 2 pounds.

3.7 Operation. The life preserver shall be buoyant and all vulcanizing, sealed areas, and cemented attachments shall remain intact without indication of material failure, separation or leakage when tested as specified in 4.5.2.1.2.

3.8 Workmanship. The life preserver shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, and durability. The life preserver shall be capable of sustaining the weight of a 247.2 pound soldier in water with the soldier's nose and mouth completely out of the water.

3.9 Size. The life preserver shall be compatible with 5th percentile female through 95th percentile male soldier as defined by MIL-STD-1472.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article.

4.3.1 First produced.

4.3.1.1 Examination. The first produced life preserver shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Tests. The first article life preserver shall be tested as specified in 4.5.2.1.1 through 4.5.2.1.3 in that order. Failure of any test shall be cause for rejection.

4.3.2 Initial production. When specified (see 6.2), the government will randomly select one or more initial production life preservers from the first 20 life preservers being produced using production tooling. The selections will be examined as specified in 4.5.1 and tested in accordance with 4.5.2.1.1, 4.5.2.1.2, and 4.5.2.1.3, in that order, to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of a first production life preserver shall not exclude the remaining life preservers from the quality conformance inspection and acceptance provisions specified in section 4.

4.3.2.1 Inspection failure. Failure of a first article production life preserver to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3.2 shall be cause for rejection of the initial production life preservers and shall be cause for refusal by the Government to continue acceptance of production life preservers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on life preservers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all life preservers accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Sampling.

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4.4.1.1 Life preservers. Sampling for examination and tests of complete life preservers shall be in accordance with MIL-STD-105, inspection level S-3.

4.4.1.2 Adhesion samples. Five representative samples of the coated side of the cloth cemented to the coated side of the cloth, and five representative samples of the coated side of the cloth cemented to the uncoated side of the cloth shall be prepared during manufacture of each lot of life preservers. Material and techniques used in the preparation of sample joints shall be the same as used for the life preservers.

4.4.2 Examination.

4.4.2.1 Individual. Each life preserver shall be examined for the critical defects specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Life preserver. Samples selected in accordance with 4.4.1.1 shall be tested as specified in 4.5.2.1.1, and 4.5.2.1.2. Failure of any test by any sample shall be cause for rejection of represented lot of life preservers.

4.4.3.2 Adhesion samples. Adhesion samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2.2. Failure of the test by any sample shall be cause for rejection of the represented lot of life preservers.

4.5 Inspection procedure.

4.5.1 Examination. Life preservers shall be examined for the following defects:

Critical

1. Materials used for life preserver air compartments not as specified (see 3.3).
2. Air compartment cemented joints not as specified (see 3.4.1).
3. Holes, tears, cuts, abrasions, or any imperfections or damage to air compartments (see 3.8).
4. Inflation valve not as specified (see 3.5).
5. Size of life preserver not as specified (see 3.9)

Major

101. Materials except those used for the fabrication of air compartments not as specified (see 3.3).

102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).

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103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the life preserver (see 3.3.4).
106. Size and dimensions not as specified (see 3.1).
107. Inside surface of life preserver not as specified (see 3.4).
108. Spacer pockets do not separate inside of life preserver into individual air compartments (see 3.4).
109. Cemented surfaces not clean and free of foreign material prior to application of adhesive (see 3.4.1).
110. Adhesive not evenly applied to cemented surfaces (see 3.4.1).
111. Entrapped air or wrinkles not removed from cemented joints (see 3.4.1).
112. Components missing or not as specified (see 3.1).
113. Stitching missing or not as specified (see 3.4.2).
114. Ends of stitching not backstitched as specified (see 2.3.1).
115. Thread breaks, skips, or run-offs not overstitched as specified (see 3.4.2).
116. Weight of life preserver exceeds two pounds (see 3.6).

Minor

201. Identification markings or instruction markings missing or not as specified (see 3.1).
202. Color not as specified (see 3.1).
203. Inside surfaces of pockets not as specified (see 3.4).
204. Application of tape used for reinforcement of cemented joints not as specified (see 3.4).
205. Workmanship not as specified (see 3.8).

4.5.2 Tests.

4.5.2.1 Life preserver.

4.5.2.1.1 Oral inflation valve and stem. With the oral inflation valve outside of the loop, grasp the valve and exert pull in the direction of the longitudinal axis of the tube until the tube is stretched a minimum of 5 inches more than its normal length. Repeat the test by swinging the tube 180 degrees and exerting pull in the opposite direction as originally applied until the tube is stretched a minimum of 5 inches more than its normal length. Any indication of separation of the tube from the life preserver or the valve from the tube shall constitute failure of this test.

4.5.2.1.2 Operation. The life preserver shall be inflated by introducing moisture-free air through the oral inflation valve to a minimum pressure of 5 pounds per square inch gage (psig) and that pressure maintained for a minimum of 5 minutes. The internal pressure shall then be adjusted to 2 psig and the life preserver totally submerged in water for a minimum of 5 minutes. An internal pressure drop equal to or greater than 0.5 psig shall constitute failure of this test. Any evidence of material failure or separation accompanied by air bubbles shall also constitute failure of this test.

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4.5.2.1.3 Cemented joints. Ten samples shall be taken at random from each life preserver and shall be tested in accordance with ASTM D 413, machine method. Five of the ten samples shall be of coated side of cloth cemented to coated side of cloth. The other five samples shall be of the coated side of cloth cemented to uncoated side of cloth. Failure of one or more samples to meet the minimum strength requirement specified in 3.4.1 shall constitute failure of this test.

4.5.2.2 Adhesion. Adhesion samples shall be tested in accordance with ASTM D 413, machine method. Failure of any test sample to meet the minimum strength requirements of 3.4.1 shall constitute failure of this test.

4.6 Inspection comparison. When initial production is specified (see 6.2), the Government may select life preservers at any time during the contract production period and subject these life preservers to the examination specified in 4.5.1 and the tests specified in 4.5.2.1.1, 4.5.2.1.2, and 4.5.2.1.3 in that order, to determine that the quality of the selected life preservers is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Life preservers will be selected at random from those which have been accepted by the Government and will not include the previously inspected initial production life preservers.

4.6.1 Inspection failure. Failure of any life preserver when inspected as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of life preservers until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.7 Inspection of packaging.

4.7.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.3 Examination. Samples selected in accordance with 4.7.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

117.	Containers not as specified for levels A or B (see 4.2.1 and 5.2.2).
118.	Life preserver not arranged in fiber box so as to prevent kinking or bending of the oral inflation tube (see 5.1.1).
119.	Each box not closed and waterproof sealed as specified for level A (see 5.1.1).
120.	Gross weight of plywood boxes exceeds 200 pounds for level A (see 5.2.1).
121.	Strapping not zinc coated for level A packing (see 5.2.1).
122.	Fiber boxes exceed the size and weight limitation for level B packing (see 5.2.2)
123.	Marking illegible, incomplete, incorrect (see 5.3).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each life preserver shall be preserved in a fiber box conforming to PPP-B-636, W5c, style RSC. The life preserver shall be arranged in the box in a manner to prevent kinking or bending of the oral inflation tube. After closure, the box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

5.1.2 Level C. Each life preserver shall be arranged and preserved as specified for level A, except the box shall be domestic class, SW, style optional, grade 125. Waterproof sealing of the box with tape shall not be required.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The life preservers, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. The life preservers preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-640, class 2, style optional; or PPP-B-636, V3c, style RSC. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. The life preservers, preserved as specified in 5.1, shall be packed as specified for level B except the box shall be class 1 or domestic, whichever is applicable, and strapping may be finish A.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. In addition, the phrase "THIS PACKAGE SHALL NOT BE OPENED UNTIL ISSUED TO THE USER" shall be marked on individual unit containers (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Life preserver design 9008 is intended for use by personnel engaged in river-crossing and amphibious operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval, and the number of units required (see 4.3.2).

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d. Time frame required for submission of first article life preserver (see 3.2)
e. Level of preservation and packing required (see 5.1 and 5.2).
f. When special marking is required (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be an initial production item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Oral inflation
Personal flotation device

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
DLA - CS

User activity:
Army - AT

Preparing activity:
Army - ME

Project 4220-A309

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-52413B(ME)	9 May 1990

3. DOCUMENT TITLE
Life Preserver, Yoke: Oral Inflation, Trapped Air Type, Design 9908

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
	(703) 664-5717 · 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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